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Engineering Mathematics

8Week [24 October - 30 October]

Quiz 8

|              |                        |
|--------------|------------------------|
| Started on   | 2023-10-30 04:06       |
| State        | Finished               |
| Completed on | 2023-10-30 04:19       |
| Time taken   | 12 mins 38 secs        |
| Grade        | 4.33 out of 5.00 (87%) |

### Question 1

Correct

Mark 1.00 out of 1.00

Using just the definition find the Laplace transform of  $f(t) = t$ .

Select one:

- ☐  $F(s) = s$
- ☐  $F(s) = \frac{1}{s}$
- ☐  $F(s) = s^2$
- ☒  $F(s) = \frac{1}{s^2}$  ✓

Your answer is correct.

The correct answer is:  $F(s) = \frac{1}{s^2}$

### Question 2

Correct

Mark 1.00 out of 1.00

Which properties we used to prove linearity of the Laplace transform?

Select one or more:

- ☒ Linearity of integration ✓
- ☐ Linearity of differentiation
- ☒ Distributivity (opening the brackets) ✓
- ☐ Commutativity (changing order)

Your answer is correct.

The correct answers are: Linearity of integration, Distributivity (opening the brackets)

Question 3

Correct

Mark 1.00 out of 1.00

Given that  $\mathcal{L}\{\cos bt\}(s) = \frac{s}{s^2+b^2}$ , use the translation property to

compute  $\mathcal{L}\{e^{at} \cos bt\}$

Select one:

☒  $\frac{s-a}{(s-a)^2+b^2}$  ✓

☐  $\frac{s+a}{(s+a)^2+b^2}$

☐  $\frac{b}{(s-a)^2+b^2}$

☐  $\frac{b}{(s+a)^2+b^2}$

Your answer is correct.

The correct answer is:  $\frac{s-a}{(s-a)^2+b^2}$

Question 4

Partially correct

Mark 0.33 out of 1.00

Among the functions below choose those function(s)  $f$  on

$[0, \infty)$  such that  $\mathcal{L}\{f\}(s) = \frac{1}{s}$ .

Select one or more:

☒  $f(t) = 1, t \geq 0$  ✓

☐  $f(t) = \begin{cases} 1, & 0 \leq t < 1 \\ 2, & t \geq 1 \end{cases}$

☐  $f(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 0 \end{cases}$

☐  $f(t) = \begin{cases} 2, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$

☐  $f(t) = \begin{cases} 1, & t \geq 0, t \neq 2, t \neq 3, t \neq 4 \\ 2, & t = 2 \\ 3, & t = 3 \\ 4, & t = 4 \end{cases}$

Your answer is partially correct.

You have correctly selected 1.

The correct answers are:  $f(t) = 1, t \geq 0$ .

$$f(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 0 \end{cases}$$

$$f(t) = \begin{cases} 1, & t \geq 0, t \neq 2, t \neq 3, t \neq 4 \\ 2, & t = 2 \\ 3, & t = 3 \\ 4, & t = 4 \end{cases}$$

Question **5**

Correct

Mark 1.00 out of 1.00

Find the inverse Laplace transform of the function  $\frac{s}{s^2+4}$ .

Select one:

- ☐  $\cos t$
- ☐  $\sin t$
- ☒  $\cos 2t$  ✓
- ☐  $\sin 2t$

Your answer is correct.

The correct answer is:  $\cos 2t$